CSC 453 Lab 2 Report: Kaitlyn Carrillo

1. **Get a Minix System:**
    a. **Approach:** Using the internet, I looked initially for QEMU because that seemed to be the best for my macOS. I installed it using homebrew and then I got confused on how to use it so I stopped. I remembered that someone said in class that VMware would work so I tried that instead.
    b. **Problems**: The VMware seemed to work great as I loaded the disk image onto it, but when I got to the lab Professor Nico said otherwise. I couldn't set the Hard disk to IDE which is what I needed.
    c. **Solutions**: The solution was to go back to QEMU with the guidenace of Professor Nico to get it started up.
    d. **Lessons**: I learned to use the Virutal Machine Professor Nico says to or else it won't work. I also gained the confidence to ask the professor when I need help because I am bad at doing that. Learning how to install a Minix machine onto my Mac was also pretty interesting.
2. **Log in :**
    a. **Approach**: The login procedure is to enter in your username at the login and then your password at the password prompt. The username I used was "kcarri14"
    b. **Problems:** I didn't encounter any problems with logging into root, but logging into the user account took some time. I didn't realize that I wasn't actually making a user account so I tried to log into nothing and of course it didn't work.
    c. **Solutions**: After a while I figured out that the minix system was actually telling me how to add a user the whole time. Figuring this out helped me create a user account so I had success!
    d. **Lessons**: In this section, I learned that reading is good for you and maybe I should do it more.
3. **Create a user account:**
    a. **Approach**: To create a new user account, you must enter in "useradd user group home-dir" where user is the username you want to use, group is the group you are in, and home-dir is the your home directory path.
    b. **Problems:** At first, I did it wrong and thought that I was doing it right, but unfortunately I wasn't. After I asked the internet how to do it and that was still wrong too and then I thought of something…Reading…and this worked because I got it to work.

c. **Solutions:** The Minix system outputs “Usage: useradd user group hom-dir”. So like any sane human would, I put in “useradd kcarri14 operator /home/kcarri14” and it worked!
d. **Lessons**: An error code can not look like an error code so make sure you know what the system is saying if it outputs dialogue.

4. **Create a Minix disk image and use it to store data:**
    a. **Approach**: Create an empty disk image file to simulate a floppy disk and configure the floppy drive into the Minix System. For QEMU, I had to create an empty disk image file of size 1.44MB. When booting up the minix system, I had to include the flag “-fda testfloppy”. To configure the floppy disk, I used “format /dev/fd0 1440” and then to make a file system, I used “mkfs.mfs /dev/fd0”. To mount the filesystem onto the floppy drive, I used “mount /dev/fd0 /mnt” to mount it into the /mnt directory.
    b. **Problems**: No problems were encountered in this section particularly because I asked Professor Nico a lot of questions that I didn’t format it wrong and have to redo it.
    c. **Solutions**: My solution is to ask Professor Nico in class. For example, am I doing this right before I mess up my minix? And he usually says yes!
    d. **Lessons**: Asking questions is amazing and I should do it more often because then my lab goes faster!
5. **Accessing your data from outside minix:**
    a. **Approach**: Using the CSL machines and Professor Nico’s minls and minget, access the filesystem that I created on the Minix server. After creating it and mounting it, I unmounted it and then put the testfloppy file onto the CSL with sftp. With the file on the CSL use ~pn-cs453/demos/minls testfloppy to list the files on a minix filesystem. This shows what files are on the filesystem and then ~pn-cs453/demos/minget to read the file.
    b. **Problems**: Somehow I lost where my testfloppy was so I kept getting an error saying that the file wasn’t a minix file system.
    c. **Solutions**: Professor Nico and I restarted the QEMU and re-put the file system on the disk and added a file onto it so we can see if the file system was working. We mounted it and then unmounted it on the QEMU. Lastly, we put the testfloppy file onto the CSL so that we could make sure the file system was on the floppy disk.
    d. **Lessons**: I learned so much here because I need to make sure that I am putting the correct files onto the CSL so that it doesnt take Professor Nico and I ten minutes to fix my problems.
6. **Clean up and shut down**

a. **Approach**: Type shutdown -h in the QEMU and then exit out by clicking the red button at the top.
b. **Problems**: No problems happened with this.
c. **Solutions**: The solution is to always have Professor Nico by your side when you need help.
d. **Lessons**: Asking questions isn’t embarrassing, that's what professor’s are for because if we didn’t need them, we wouldn’t have them.